LDSC Estimator Variant Comparison

All six LDSC variants (free/fixed intercept, weighted/unweighted, and two-step/single-step where the intercept is free) compared together, under real chr1/chr22 LD structure

Executive Summary

Every simulation in this project's realistic-LD pipeline saves six LDSC heritability estimator variants per replicate, varying two binary choices: intercept (free vs. fixed at 1) and SNP weighting (the standard LDSC weights vs. uniform weights). A third choice, the two-step procedure (which first estimates the intercept excluding high-chi-square SNPs, then re-estimates the slope with that intercept held fixed) vs. a single joint regression pass with both parameters free, only applies when the intercept is free - LDSC's own implementation does not permit combining two-step estimation with a fixed intercept, since there is no intercept left to estimate in a first step. So the six variants are really: two free-intercept/two-step variants (weighted and unweighted), two free-intercept/single-step variants (weighted and unweighted), and two fixed-intercept variants (weighted and unweighted, each necessarily a single joint regression pass). This report compares all six together, across every scale (n=500 through n=10000 on chr1, n=500 through n=5000 on chr22 - chr22 has not yet been run at n=10000), chromosome (chr1, chr22), and the project's four core scenarios (reference panel present or not, stratification vector f shared with the reference panel or drawn independently), for a total of 848 simulated datasets (100 replicates each).

Headline finding: in three of the four scenarios (no reference panel with either f-sharing choice, and a reference panel with *shared* f), all six variants stay within roughly a factor of 2 of each other and of h2_pop across the F_ST range shown here (0-0.1; per reviewer feedback, wider F_ST values are no longer the focus). But in the pathological scenario identified in the companion report - a reference panel whose stratification vector f is *mismatched* with the main sample - the six variants **split into two completely different failure modes**, and the split is not between two-step and single-step. The two variants that **collapse** as F_ST and scale grow (down to ~0.02-0.03 by F_ST=0.1 at n=5000, roughly 7-10% of h2_pop) are specifically the two with a *free intercept estimated in a single joint pass* (no two-step, no fixed intercept). Every other variant - the two free-intercept/two-step variants and both fixed-intercept variants - instead **inflates** with F_ST and scale, reaching up to 1.7 (roughly 5.8× h2_pop) by F_ST=0.1 at the largest scale, with SNP weighting controlling how far it inflates (weighted variants reach up to ~1.7; their unweighted counterparts stay below ~0.65).

This means the real driver of collapse-vs-inflation is whether the intercept is *stabilized* in some way - either fixed outright at 1, or estimated robustly first via the two-step procedure - before the slope is fit. Only when the intercept is both free *and* fit jointly with the slope in one pass does the regression have an unconstrained degree of freedom that can absorb the f-mismatch signal, driving the slope (and hence h2) toward zero. Fixing the intercept and using two-step are two different ways of preventing that collapse, not two points on the same axis - and SNP weighting is a largely separate, secondary axis that scales the degree of inflation among the non-collapsing variants.

Methods

See the companion report (realistic_LD_sharedf_vs_indepf_summary.pdf) for full simulation methodology. The six LDSC variants compared here, all computed from the same per-replicate data (same X, y, LD scores):

- **h2_ldsc_reg**: free intercept, weighted, two-step
- **h2_ldsc_fixed**: fixed intercept, weighted
- **h2_ldsc_fixed_nw**: fixed intercept, unweighted
- **h2_ldsc_nw**: free intercept, unweighted, two-step
- **h2_ldsc_nts**: free intercept, weighted, single-step
- **h2_ldsc_nwts**: free intercept, unweighted, single-step

All panels use a linear y-axis. The dark dash-dot line is h2_pop=0.3, the true simulated genetic heritability. The black dotted line is the theoretical total h2 (h2_pop + σ_s^2/C_f, C_f = mean(f^2/(1+f^2))), using that scenario's own per-replicate f draws - the same formula validated against the project's R implementation in the companion report. Note that in the pathological reference-panel/independent-f scenario (within the F_ST 0-0.1 range shown here), the value range spans roughly 0.02 to 1.7; on a linear axis the two free-intercept/single-step variants (which shrink toward zero as F_ST and scale grow) can appear visually compressed against the baseline next to the other four variants (which grow well above h2_pop).

Summary: All Six Estimators at F_ST=0.1, Reference Panel + Independent f

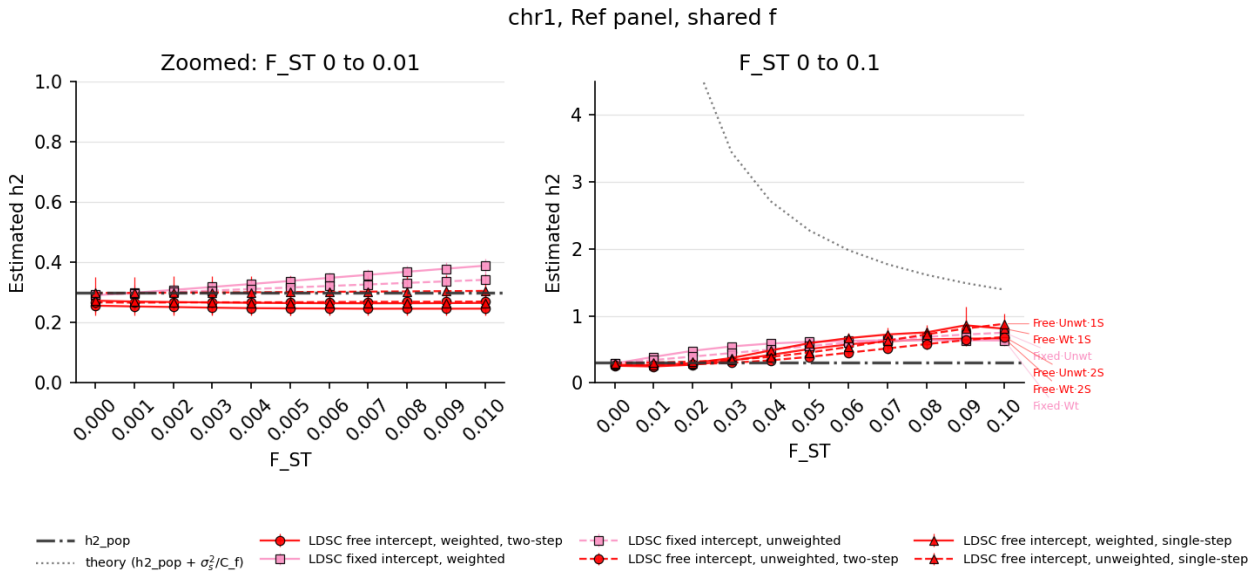
Mean h2 estimate (100 replicates) for each of the six LDSC variants at F_ST=0.1, the upper edge of the F_ST range this report now focuses on (per reviewer feedback), in the scenario where the divergence is largest: a reference panel whose f does not match the main sample's. Includes the new chr1 n=10000/m=20000 row - chr22 has not been run at that scale.

Chrom	n	Free/Wt/2step	Fixed/Wt	Fixed/Unwt	Free/Unwt/2step	Free/Wt/1step	Free/Unwt/1step
chr1	500	0.284	0.960	0.547	0.236	0.166	0.171
chr1	1000	0.438	1.123	0.581	0.282	0.097	0.064
chr1	2000	0.735	1.340	0.578	0.372	0.054	0.024
chr1	5000	1.325	1.743	0.646	0.540	0.030	0.021
chr1	10000	1.774	2.060	0.658	0.600	0.006	0.002
chr22	500	0.285	1.061	0.582	0.213	0.164	0.115
chr22	1000	0.436	1.132	0.563	0.287	0.138	0.115
chr22	2000	0.752	1.380	0.578	0.381	0.076	0.046
chr22	5000	1.283	1.692	0.643	0.539	0.026	0.031

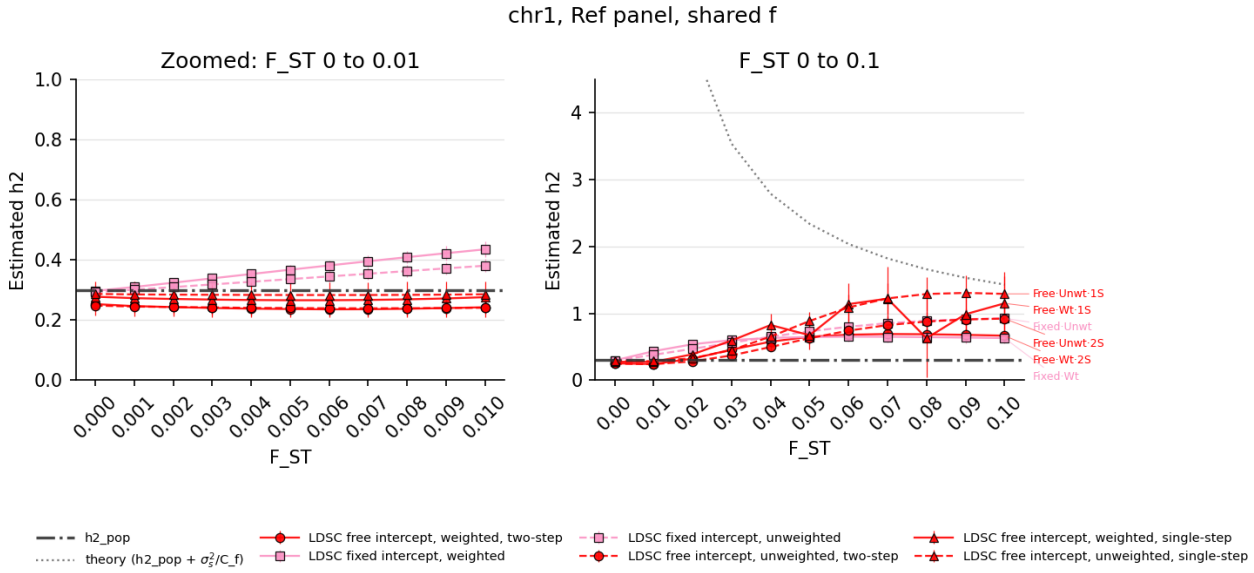
Figures: Chromosome 1

Each figure overlays all six LDSC variants (mean h2 ±95% CI) vs. F_ST, with the theoretical line overlaid (dotted). Color encodes intercept type (red=free intercept, pink=fixed intercept); linestyle encodes SNP weighting (solid=weighted, dashed=unweighted); marker shape encodes estimation mode (circle=two-step, triangle=single-step, square=fixed-intercept, which has no re-estimation step). Left panel zooms into F_ST 0-0.01; right panel shows F_ST 0-0.1.

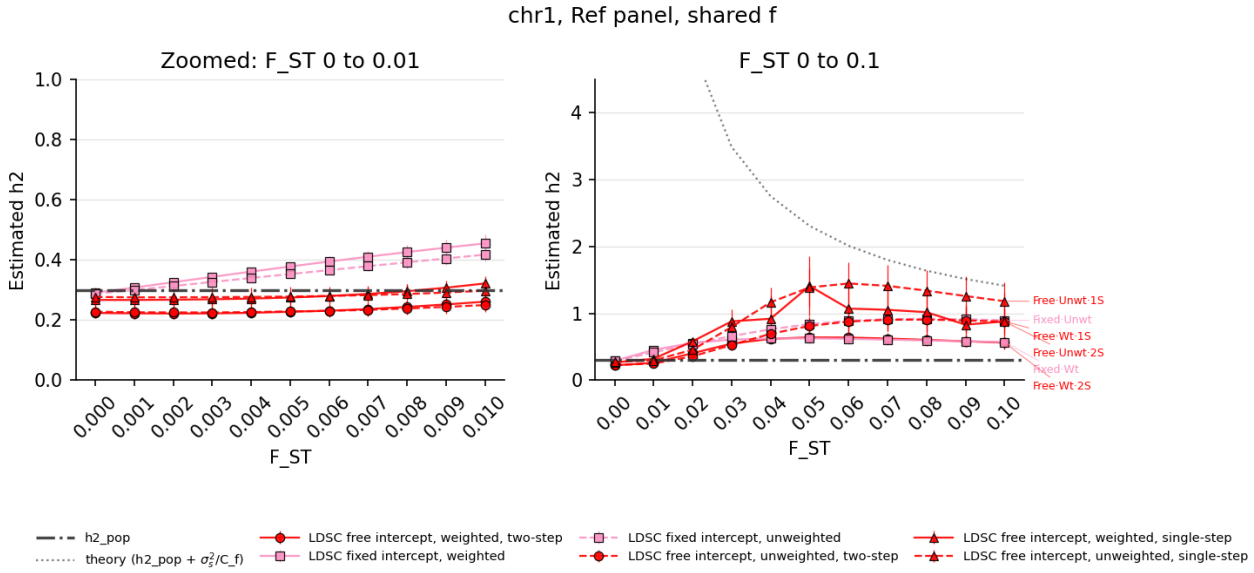
Ref panel, shared f



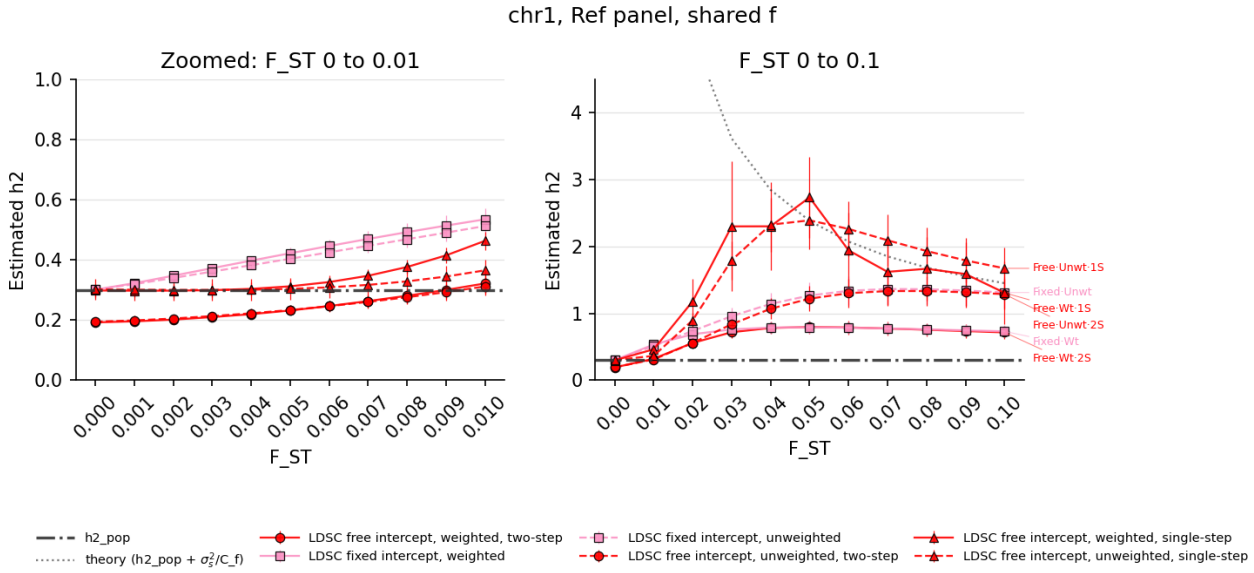
chr1, n=500, m=1000 — Ref panel, shared f



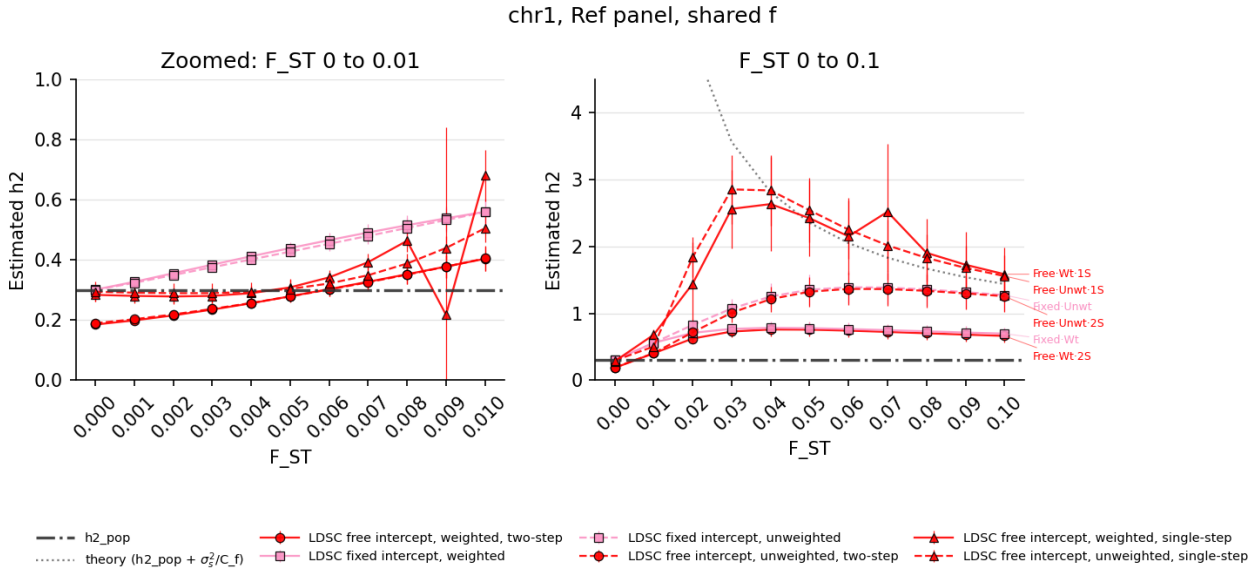
chr1, n=1000, m=2000 — Ref panel, shared f



chr1, n=2000, m=4000 — Ref panel, shared f

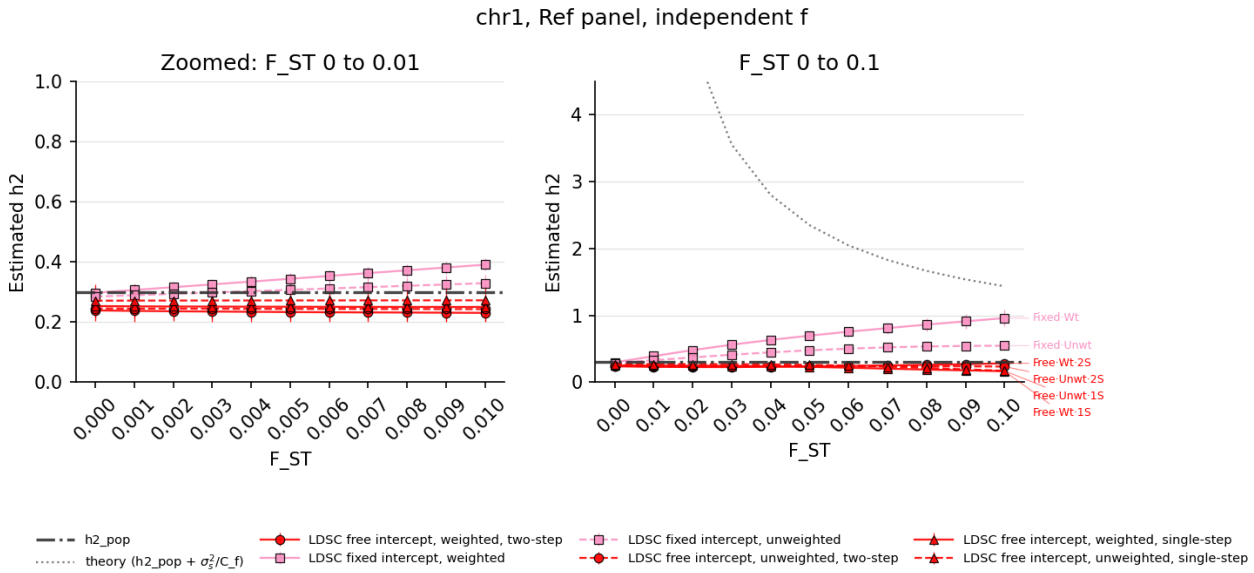


chr1, n=5000, m=10000 — Ref panel, shared f

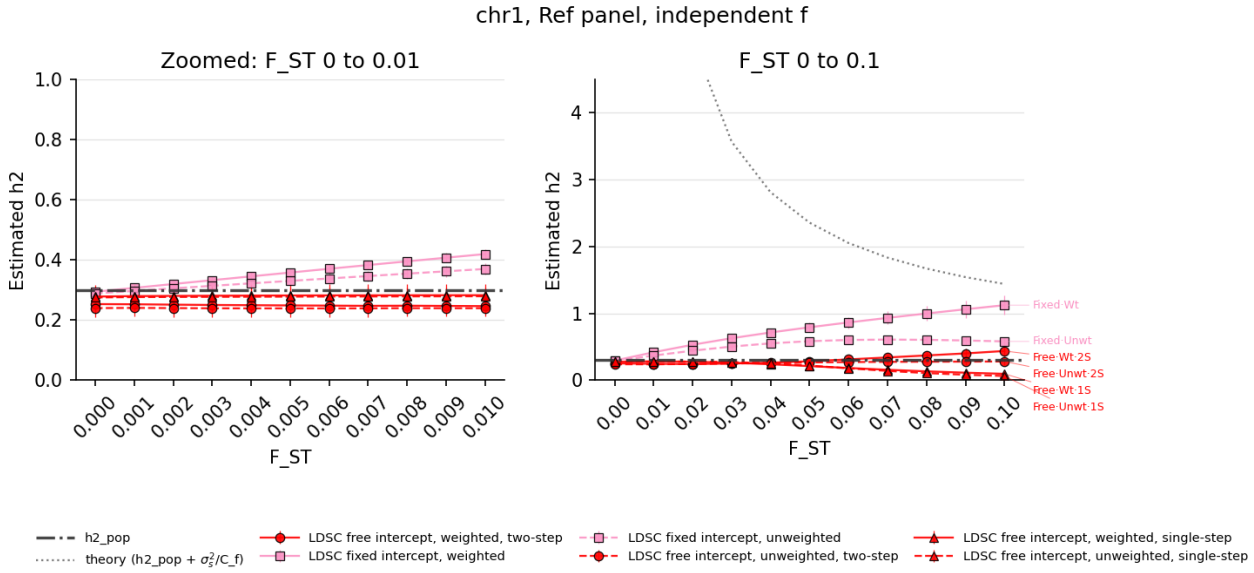


chr1, n=10000, m=20000 — Ref panel, shared f (new)

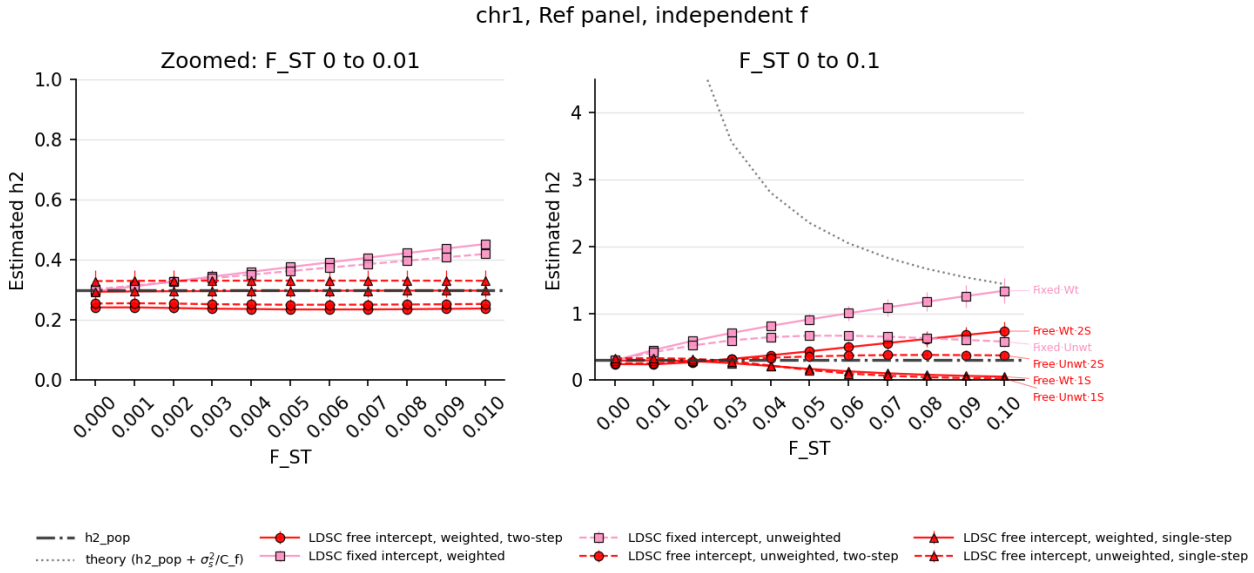
Ref panel, independent f



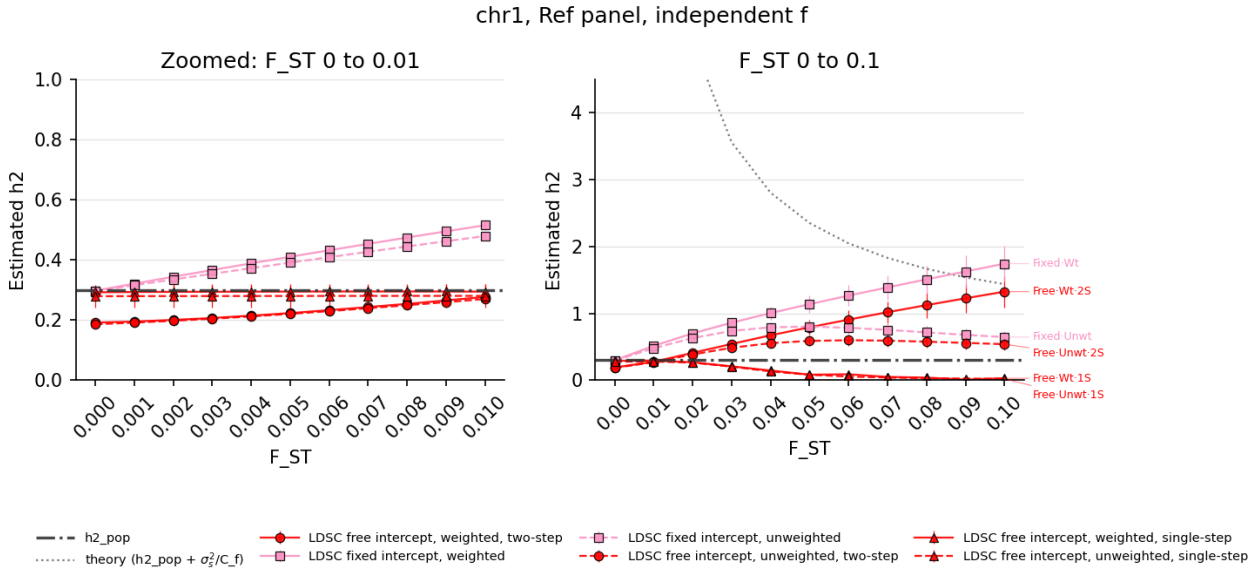
chr1, n=500, m=1000 — Ref panel, independent f



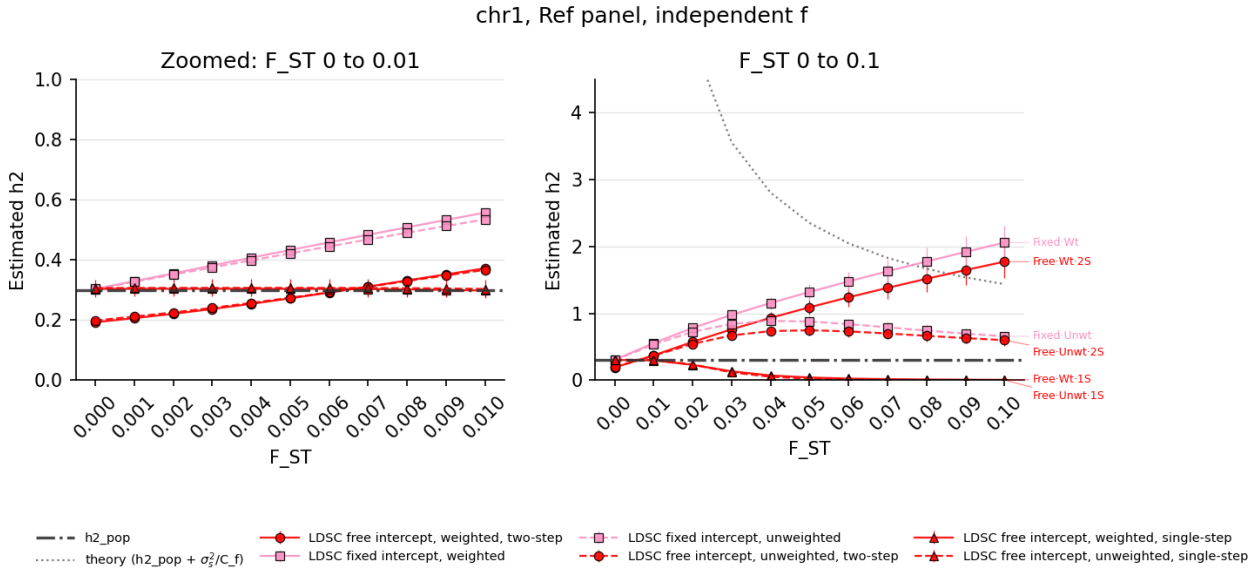
chr1, n=1000, m=2000 — Ref panel, independent f



chr1, n=2000, m=4000 — Ref panel, independent f

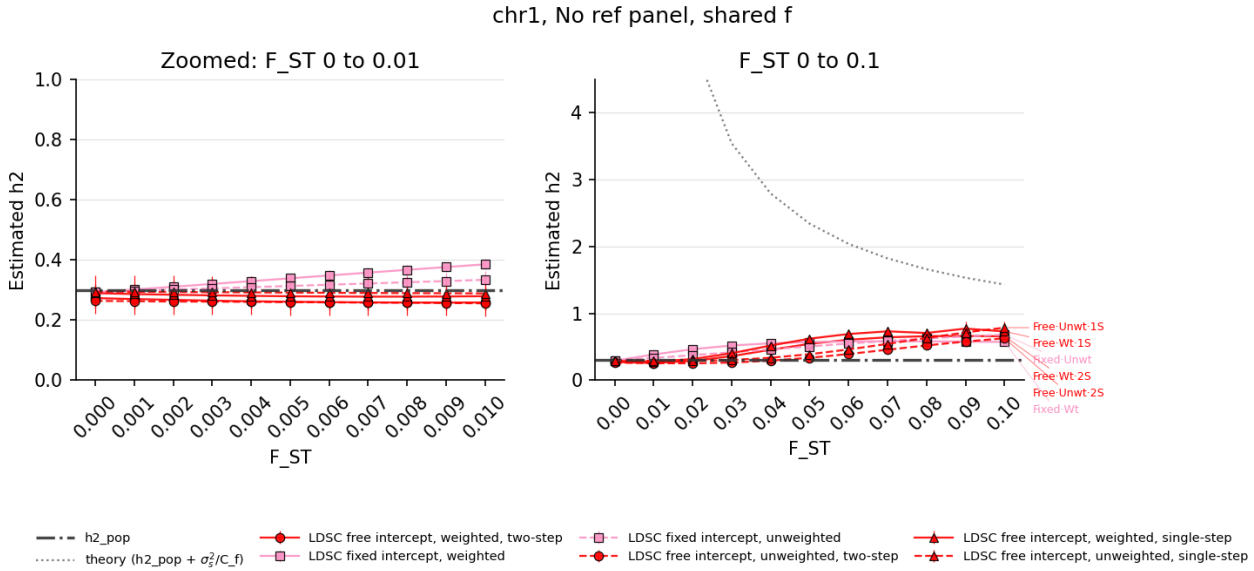


chr1, n=5000, m=10000 — Ref panel, independent f

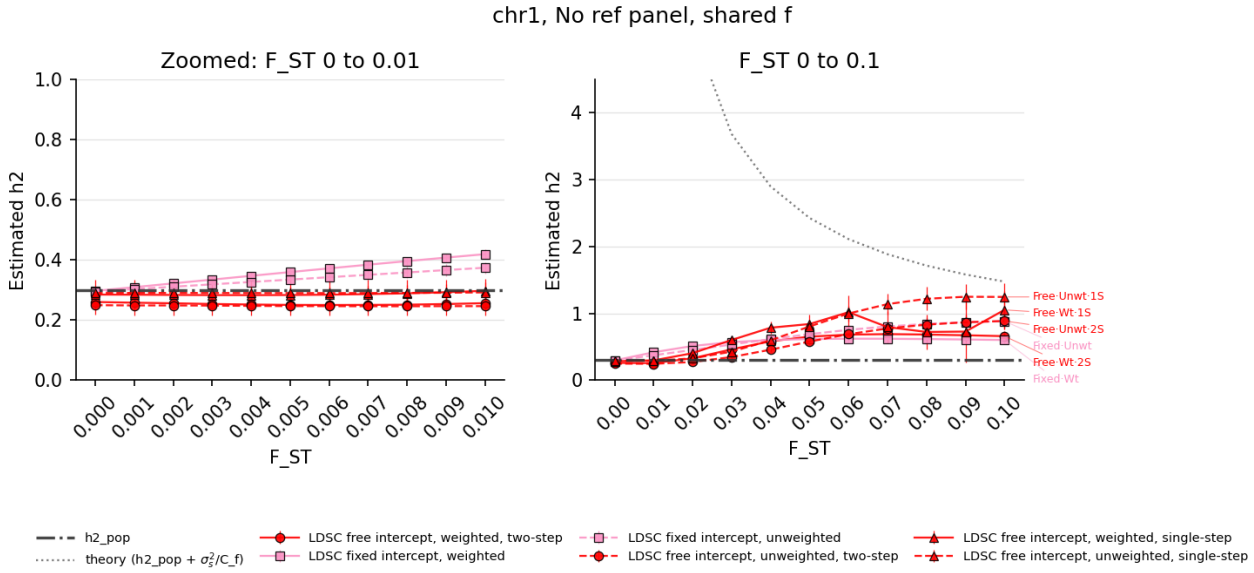


chr1, n=10000, m=20000 — Ref panel, independent f (new)

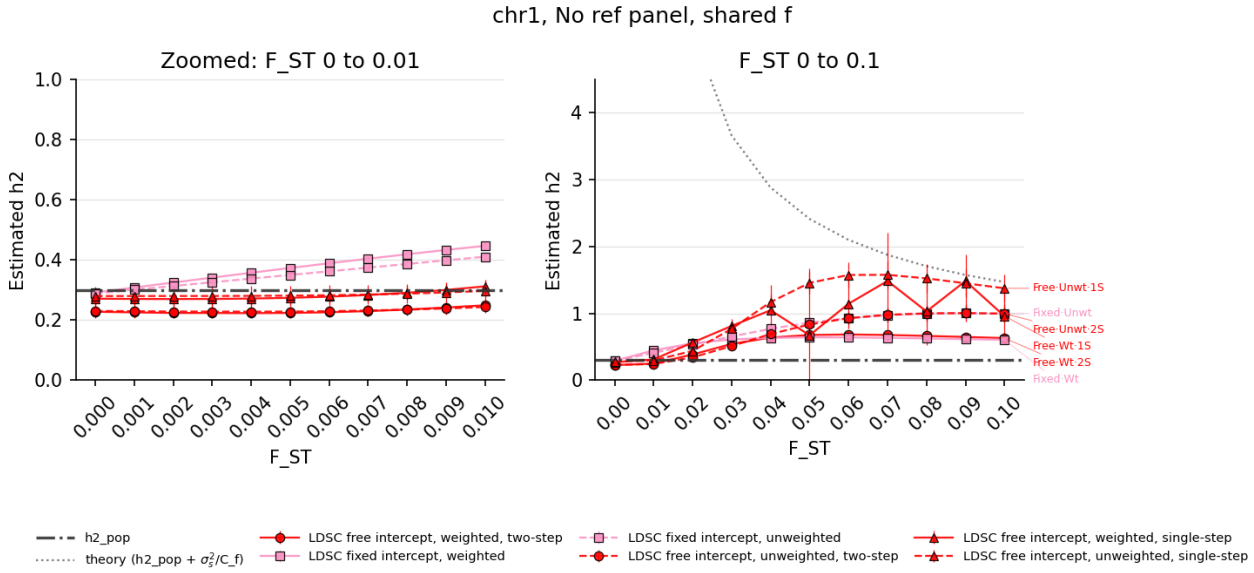
No ref panel, shared f



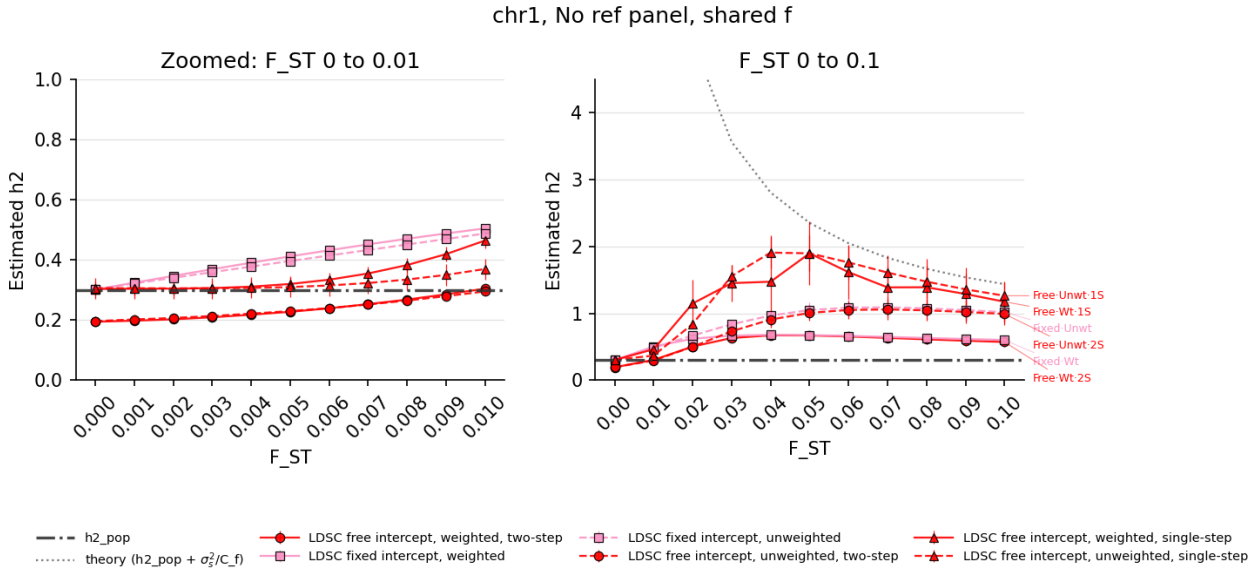
chr1, n=500, m=1000 — No ref panel, shared f



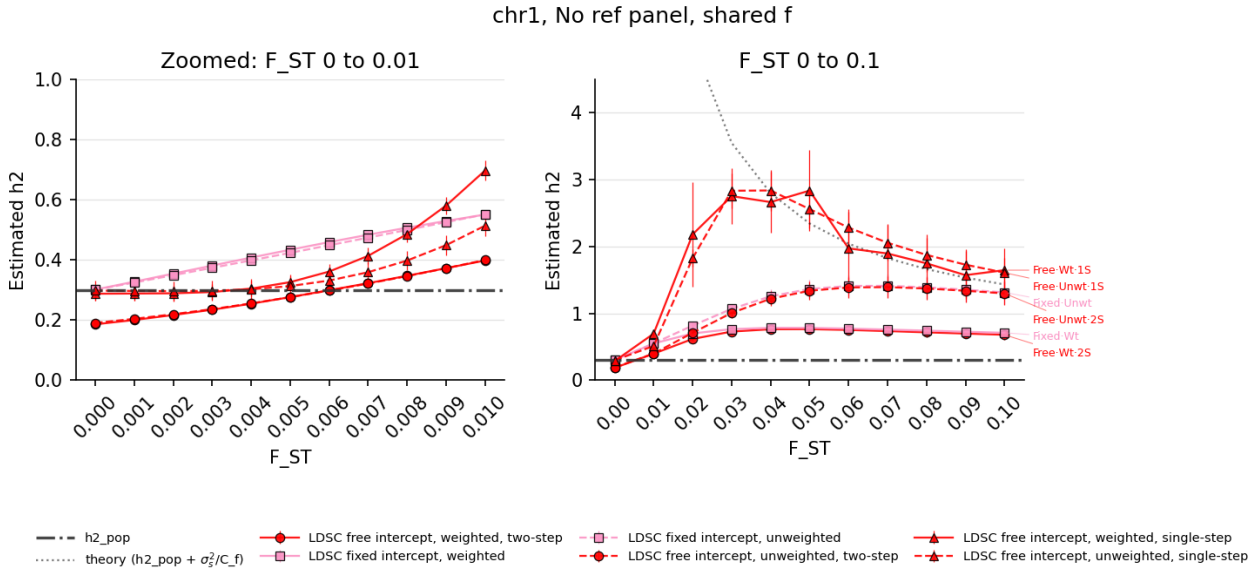
chr1, n=1000, m=2000 — No ref panel, shared f



chr1, n=2000, m=4000 — No ref panel, shared f

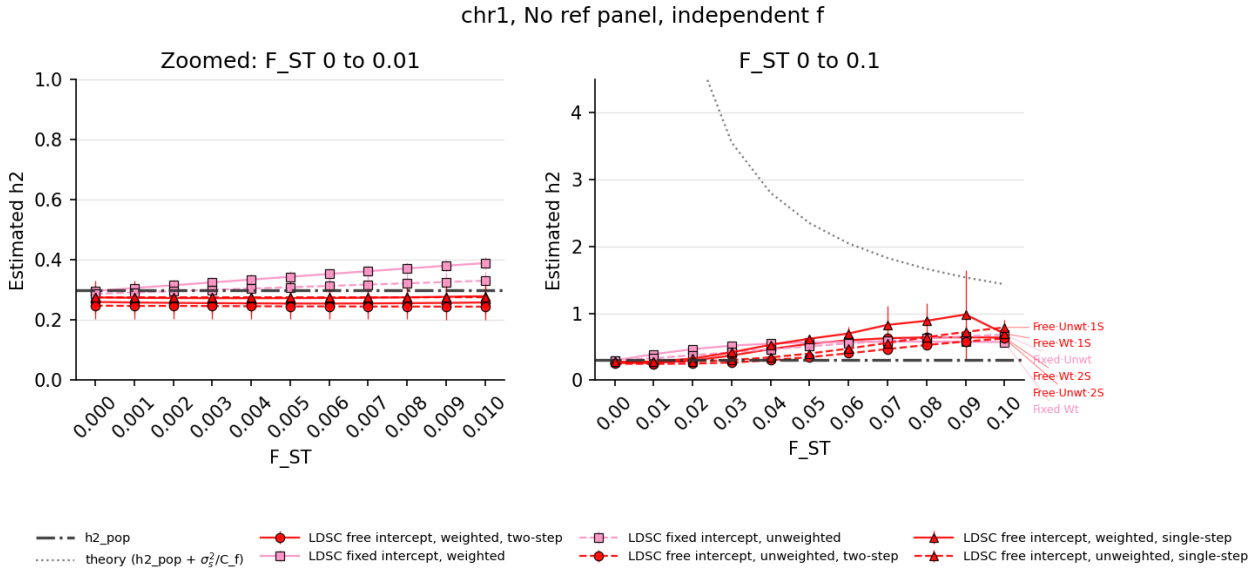


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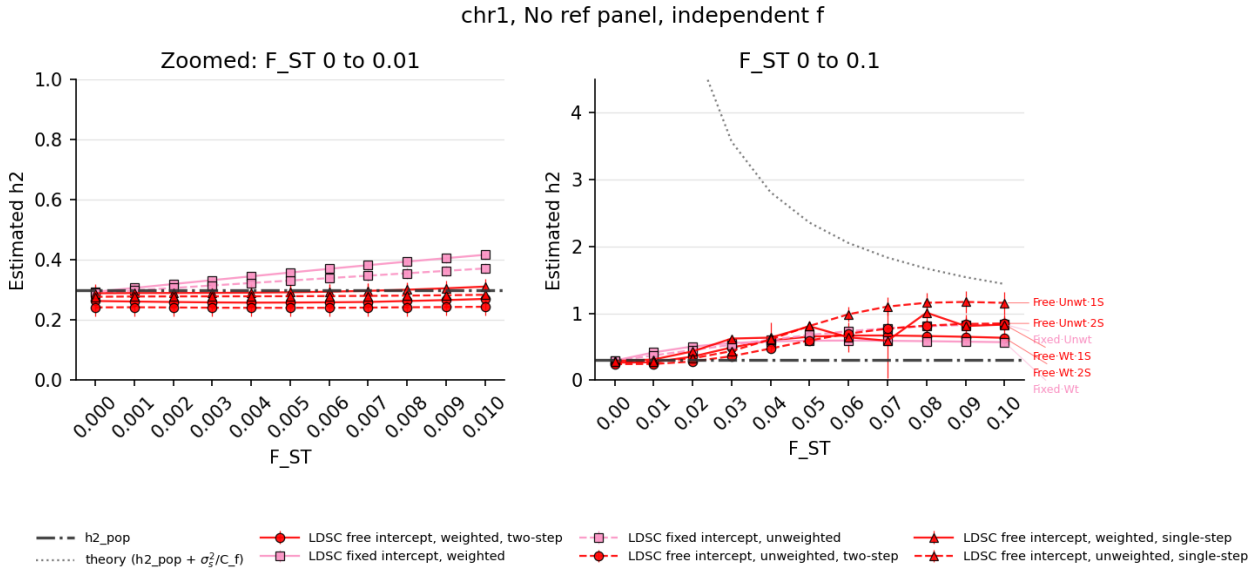


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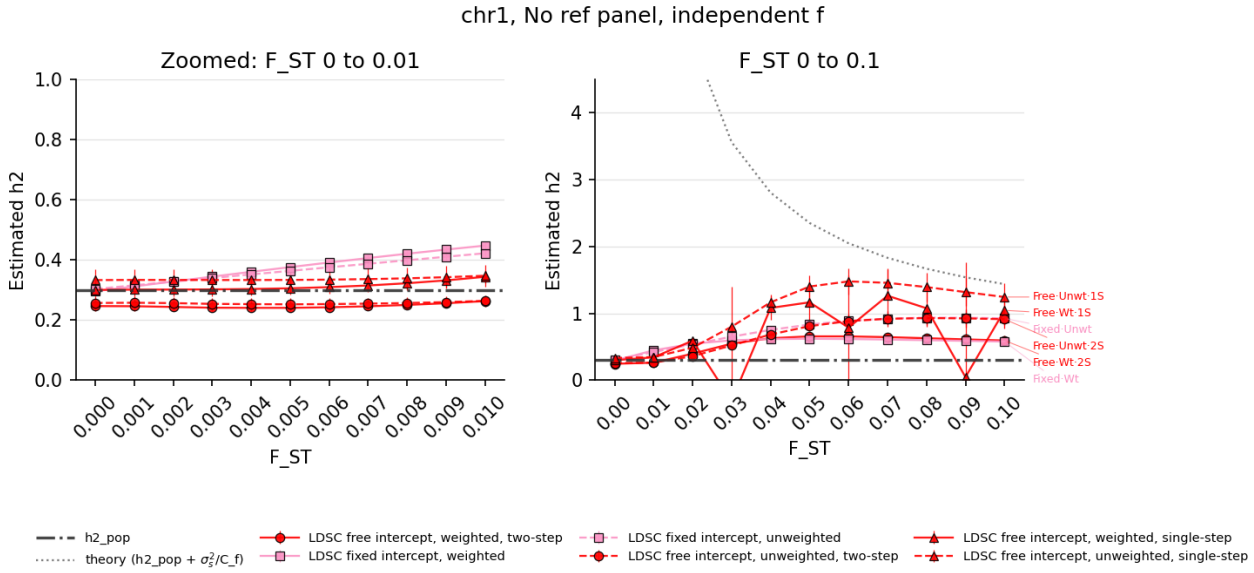
No ref panel, independent f



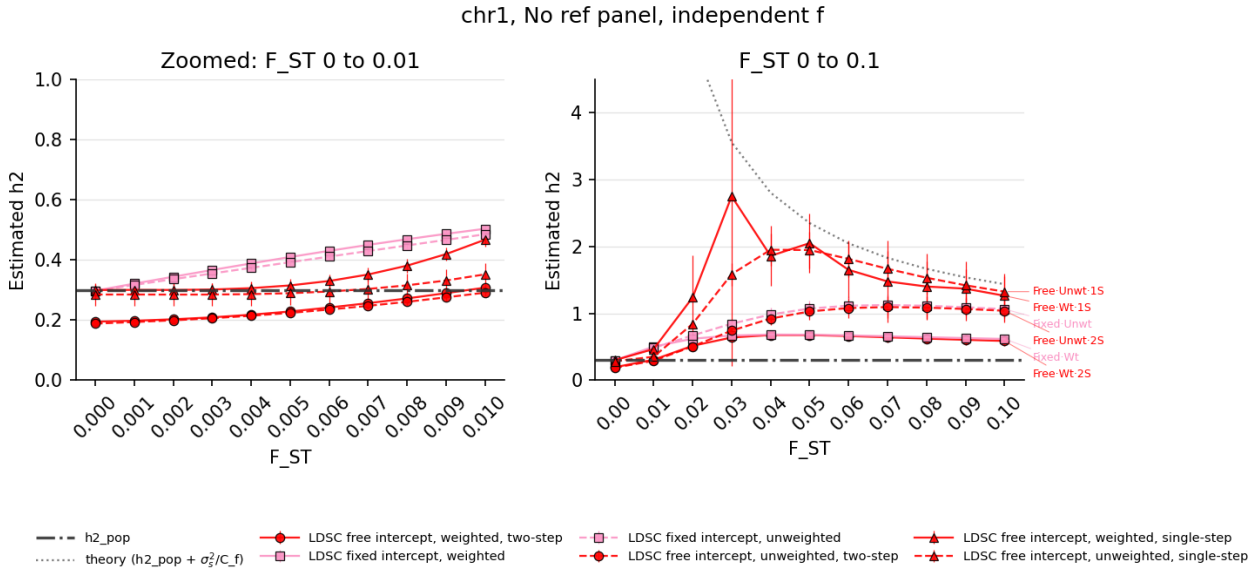
chr1, n=500, m=1000 — No ref panel, independent f



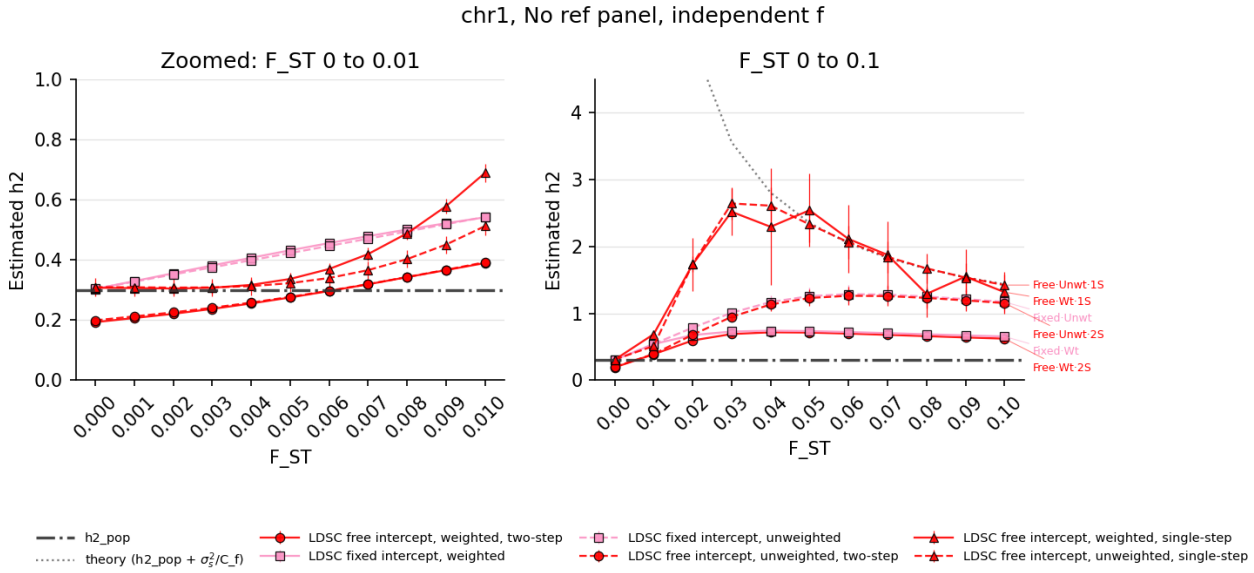
chr1, n=1000, m=2000 — No ref panel, independent f



chr1, n=2000, m=4000 — No ref panel, independent f



chr1, n=5000, m=10000 — No ref panel, independent f

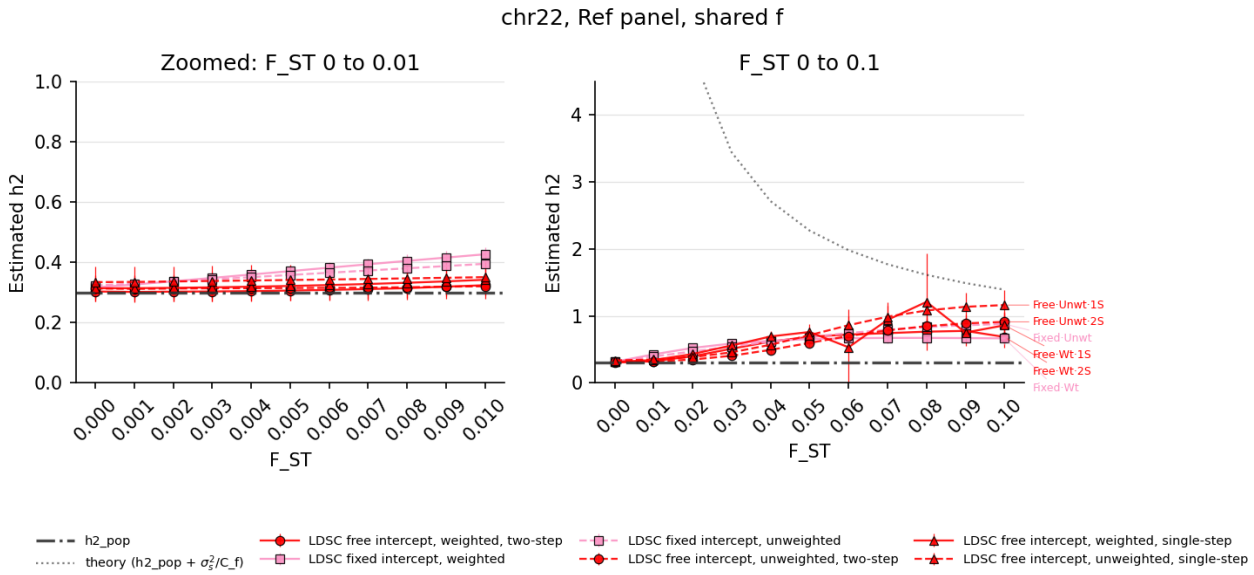


chr1, n=10000, m=20000 — No ref panel, independent f (new)

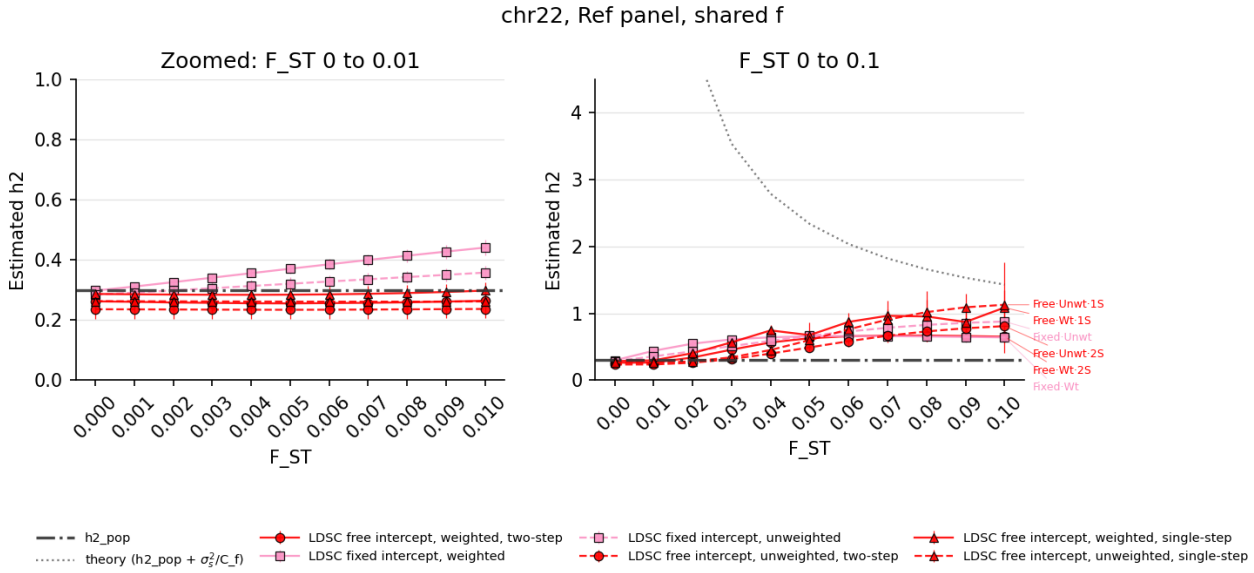
Figures: Chromosome 22

Each figure overlays all six LDSC variants (mean h2 ±95% CI) vs. F_ST, with the theoretical line overlaid (dotted). Color encodes intercept type (red=free intercept, pink=fixed intercept); linestyle encodes SNP weighting (solid=weighted, dashed=unweighted); marker shape encodes estimation mode (circle=two-step, triangle=single-step, square=fixed-intercept, which has no re-estimation step). Left panel zooms into F_ST 0-0.01; right panel shows F_ST 0-0.1.

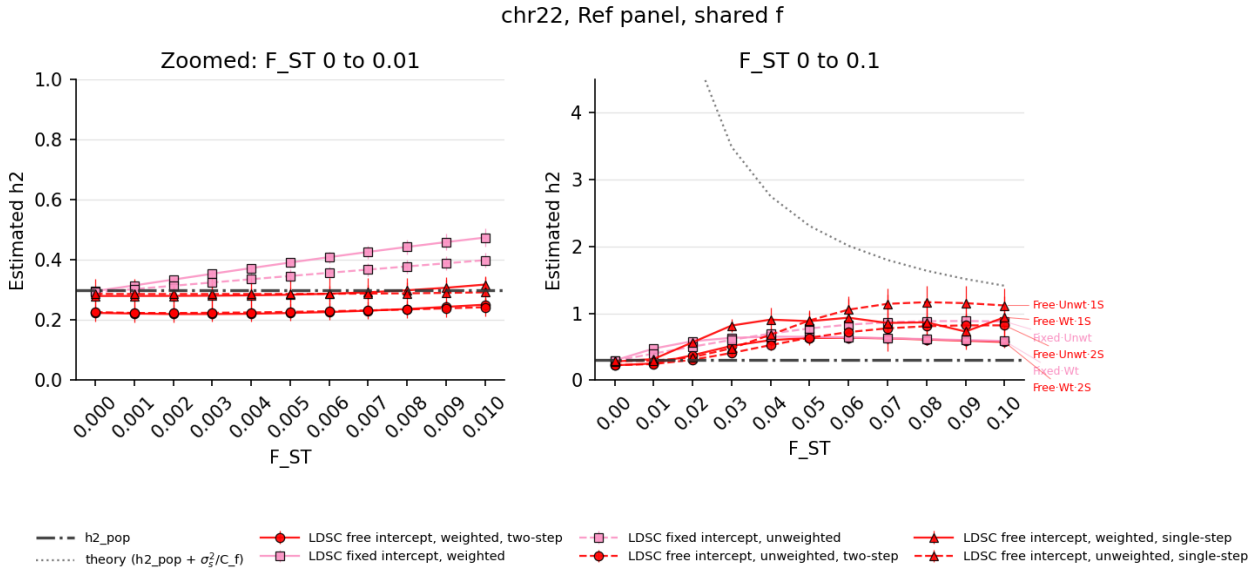
Ref panel, shared f



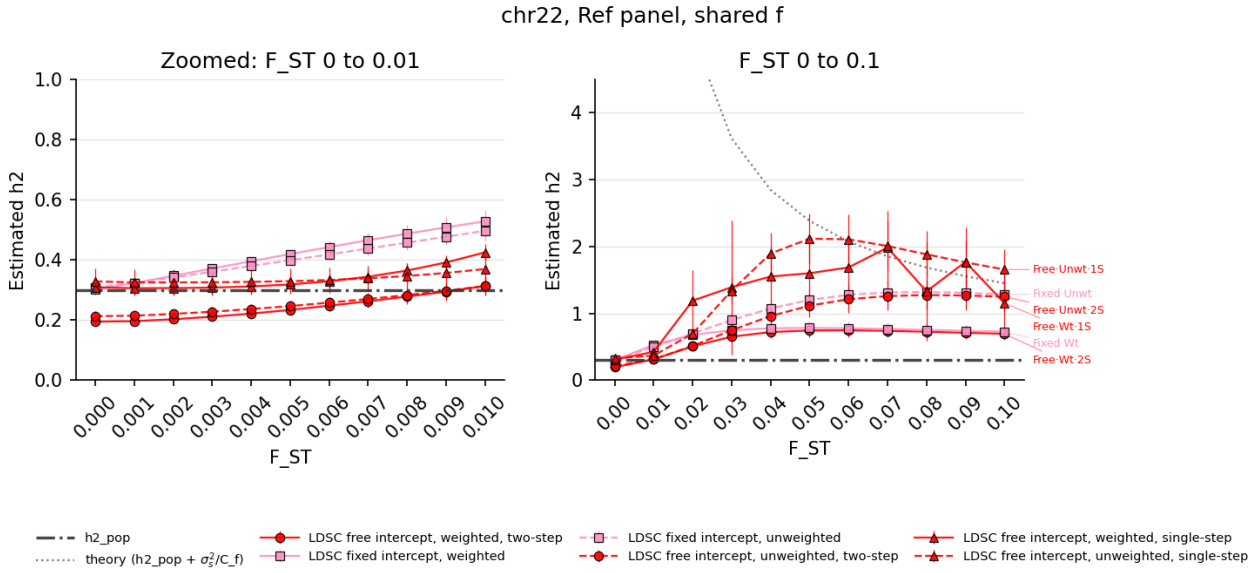
chr22, n=500, m=1000 — Ref panel, shared f



chr22, n=1000, m=2000 — Ref panel, shared f

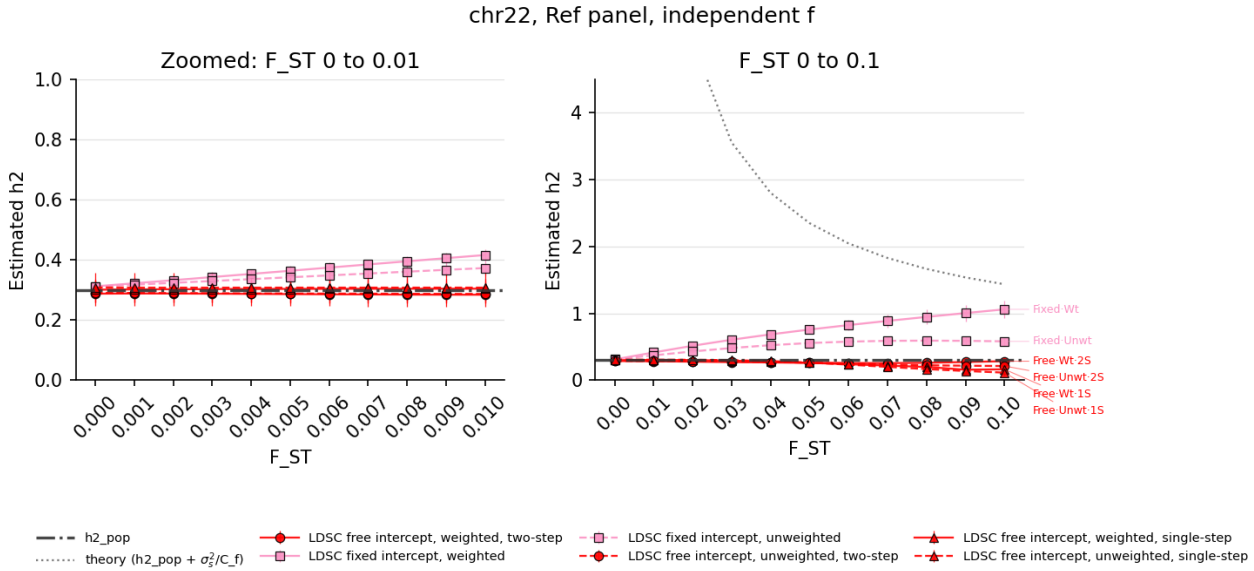


chr22, n=2000, m=4000 — Ref panel, shared f

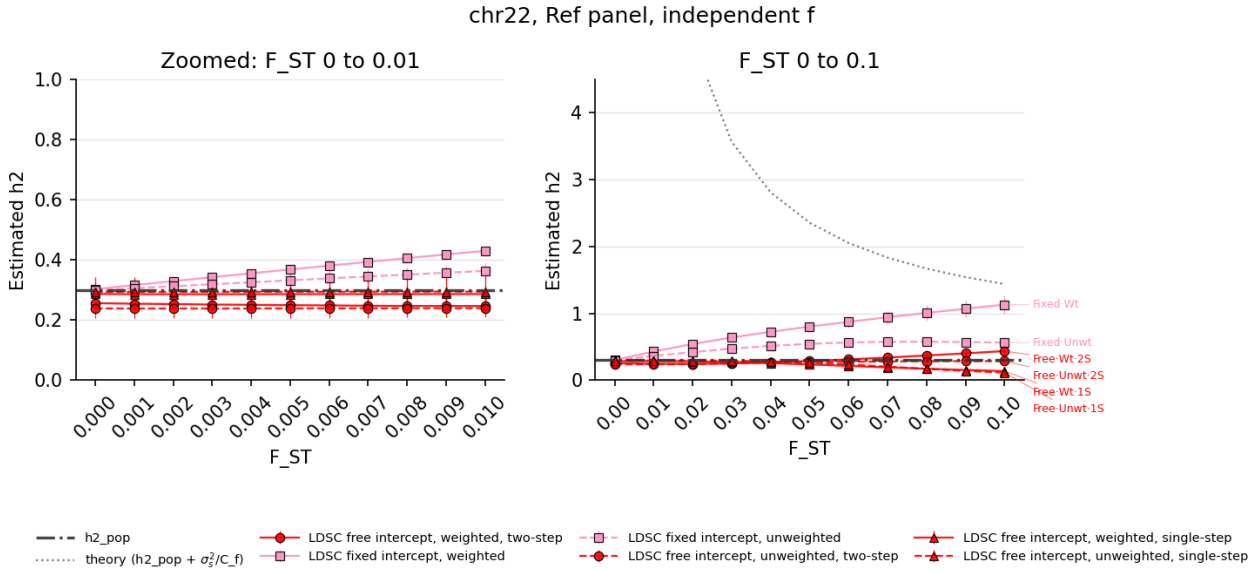


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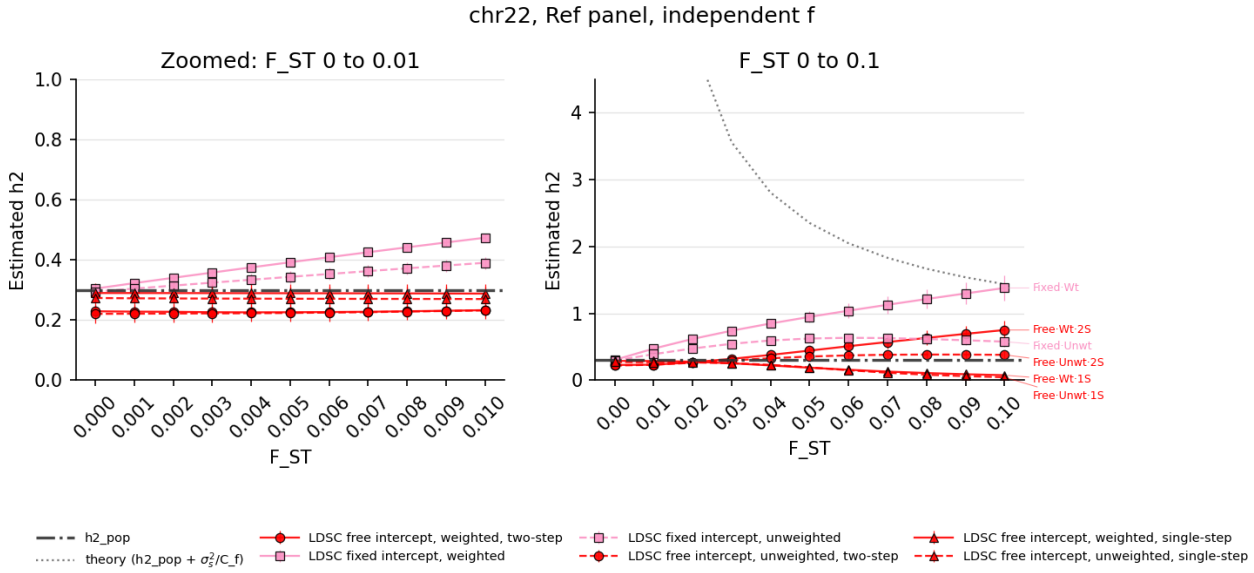
Ref panel, independent f



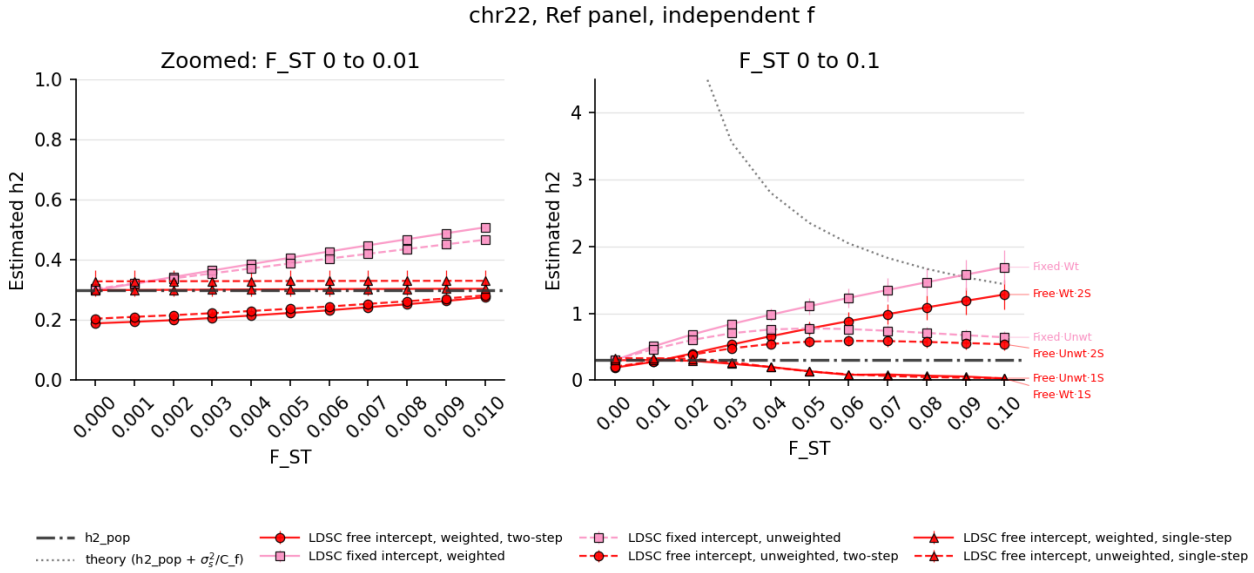
chr22, n=500, m=1000 — Ref panel, independent f



chr22, n=1000, m=2000 — Ref panel, independent f

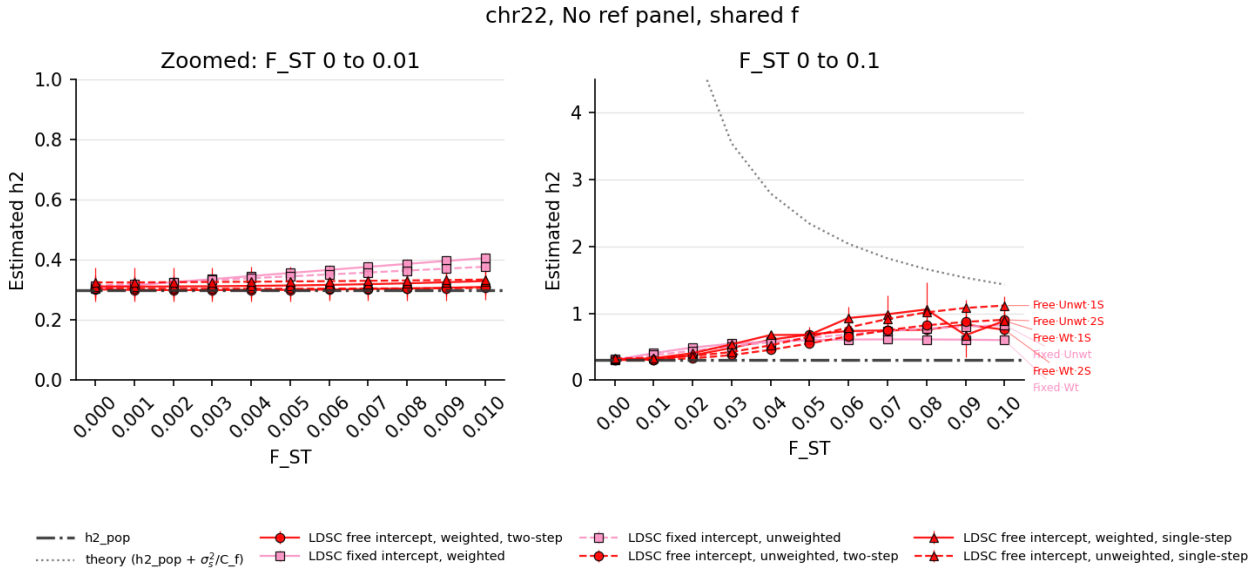


chr22, n=2000, m=4000 — Ref panel, independent f

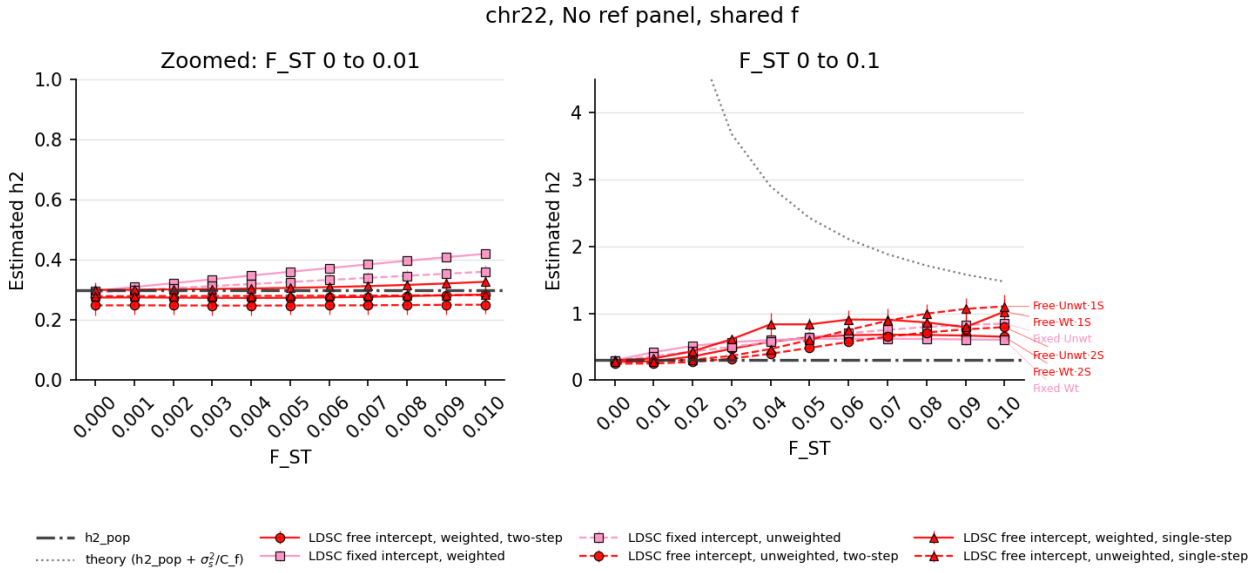


chr22, n=5000, m=10000 — Ref panel, independent f

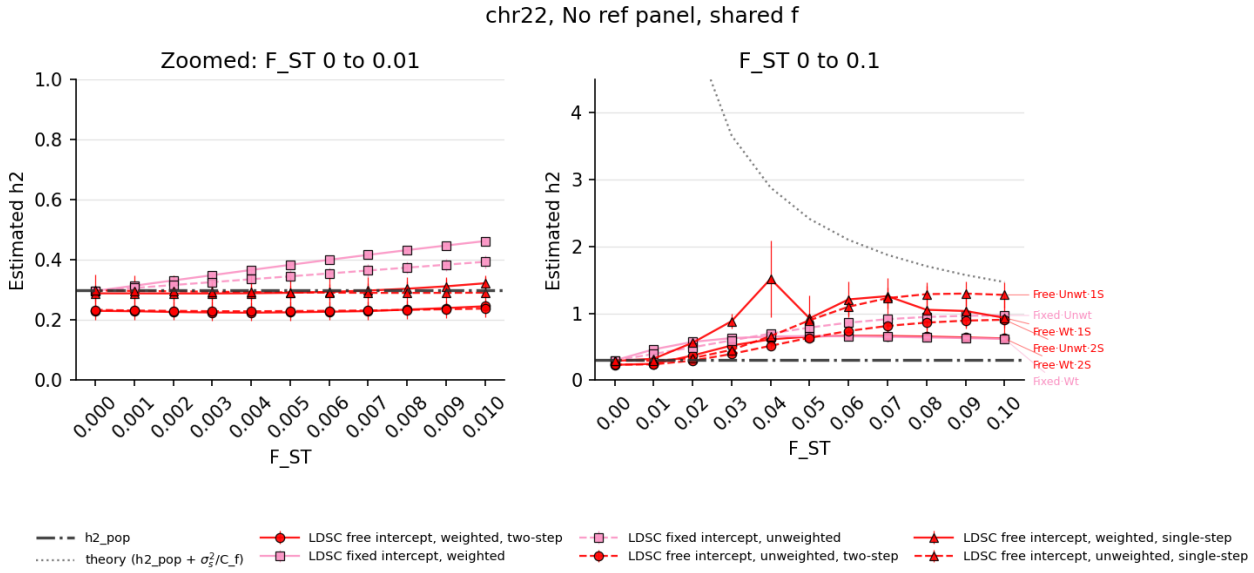
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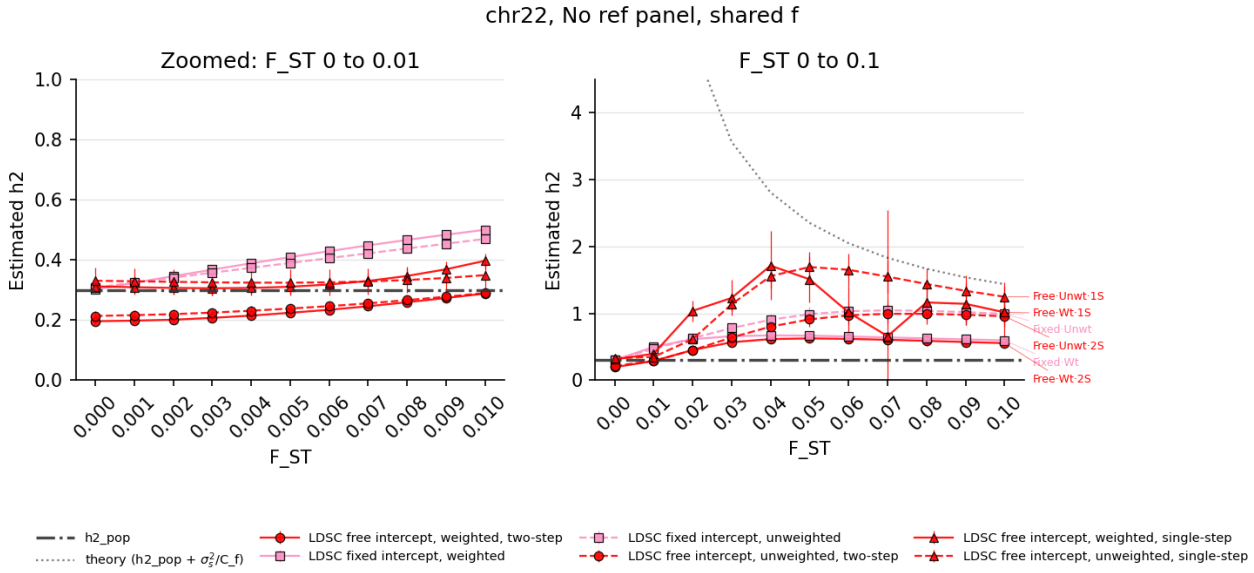
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chr22, n=1000, m=2000 — No ref panel, shared f

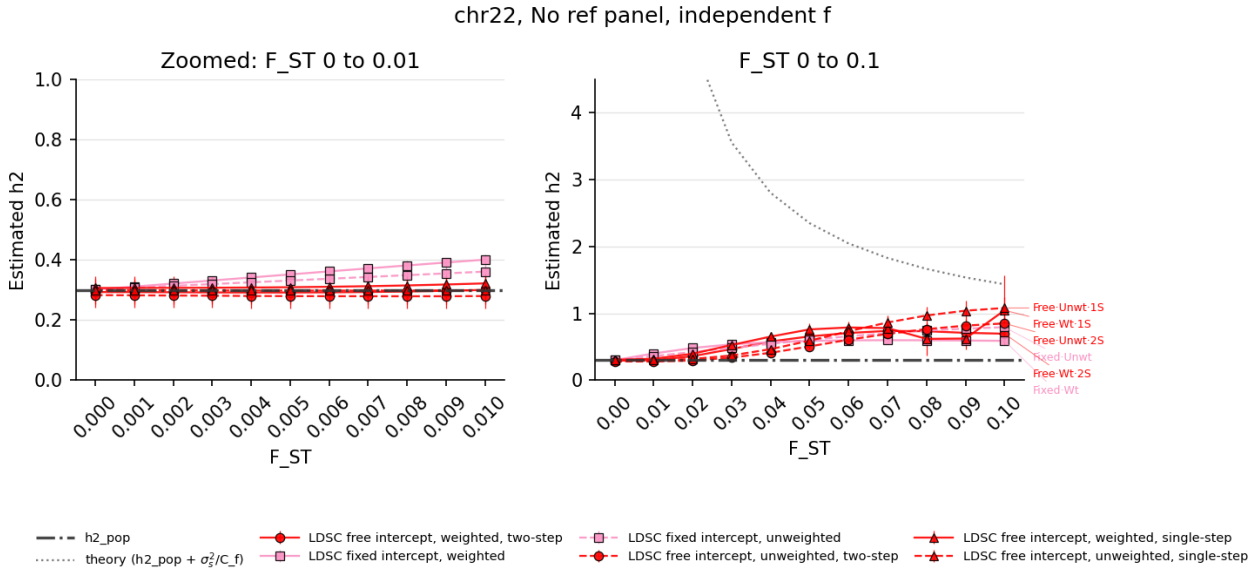


chr22, n=2000, m=4000 — No ref panel, shared f

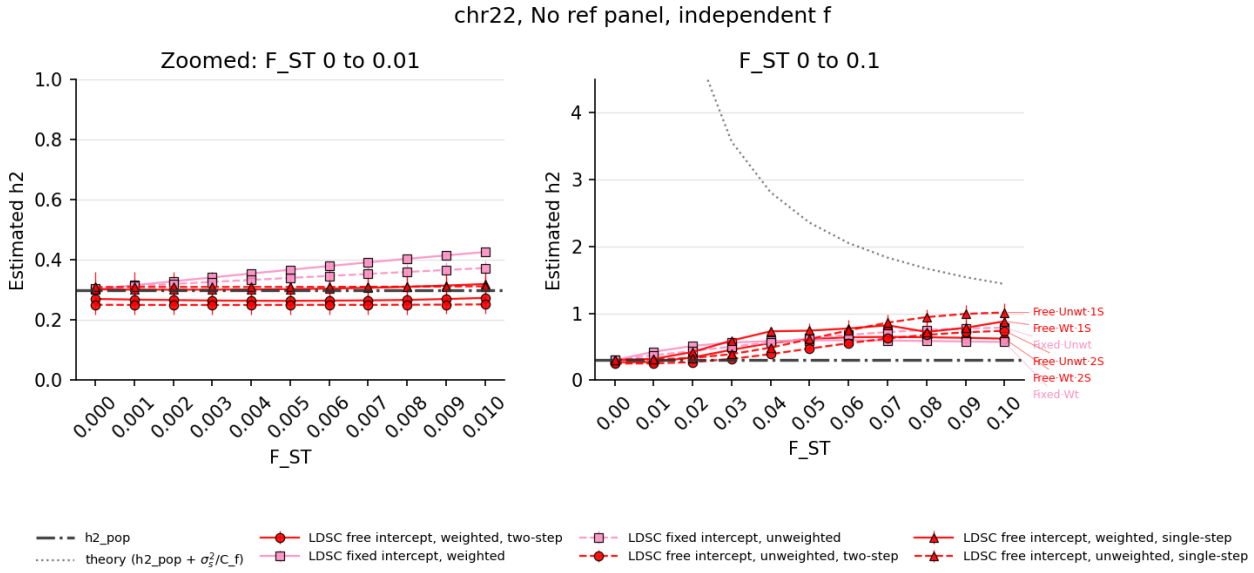


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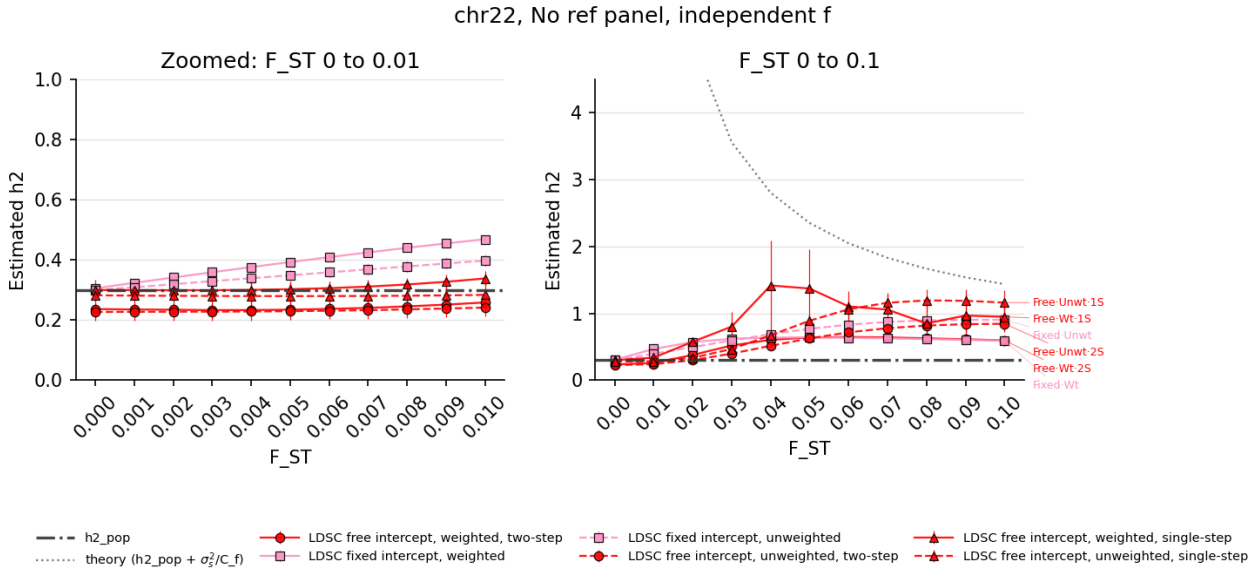
No ref panel, independent f



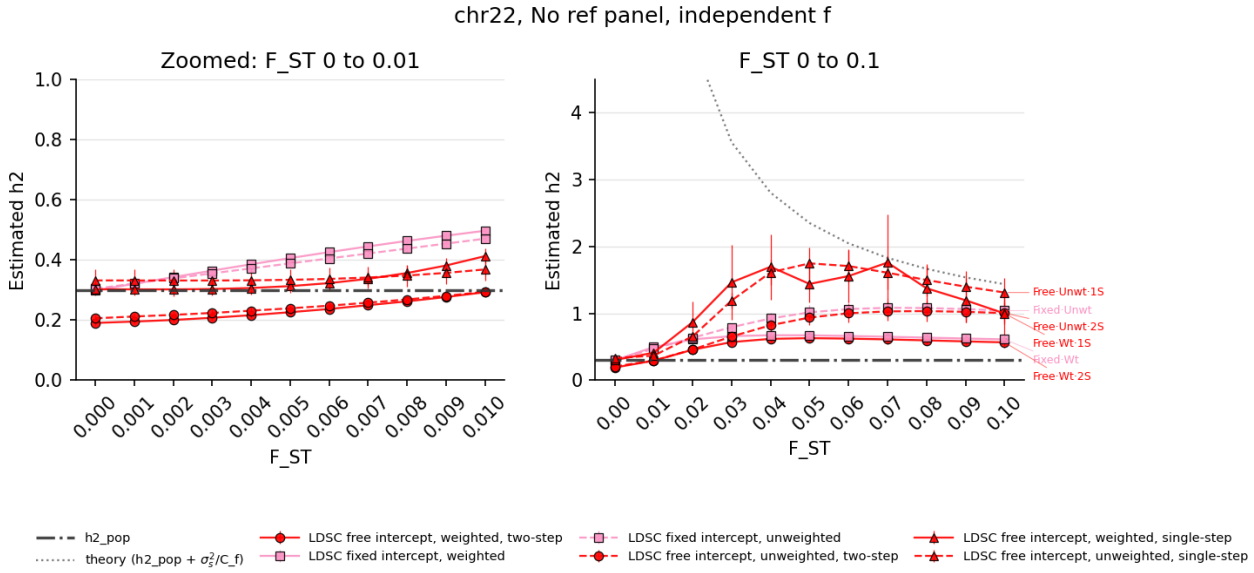
chr22, n=500, m=1000 — No ref panel, independent f



chr22, n=1000, m=2000 — No ref panel, independent f



chr22, n=2000, m=4000 — No ref panel, independent f



chr22, n=5000, m=10000 — No ref panel, independent f

Discussion

Correction to an earlier draft of this report: an initial version of this analysis mislabeled the two fixed-intercept variants as "two-step" and attributed the collapse-vs-inflation split to the two-step/single-step choice itself. LDSC's two-step procedure is only defined when the intercept is free - the project's own LDSC implementation (ldsc_barebones.do_ldsc_regression) raises an error if two-step and a fixed intercept are both requested, and the fixed-intercept driver calls never pass a twostep argument, so those two variants actually run as a single joint regression pass with the intercept held at 1. The real split, confirmed below, is between "intercept stabilized" (fixed, or free with two-step) and "intercept free and jointly fit in one pass" - not between two-step and single-step as such.

Why the free-intercept/single-step variants collapse: with the intercept both free and estimated jointly with the slope in one regression pass, the regression has an unconstrained degree of freedom that can absorb the reference panel's f-mismatch signal through the intercept term instead of the slope, driving the fitted slope (and hence h2) toward zero. The other four variants each remove that degree of freedom one way or another: two-step estimates the intercept first (on a chi-square-filtered SNP subset) and holds it fixed before fitting the slope; a literally fixed intercept removes the degree of freedom outright. Either safeguard is enough to prevent collapse - only the combination of no safeguard at all produces it.

Weighting is a separate, secondary axis: among the four non-collapsing variants, SNP weighting controls the degree of inflation almost independently of the intercept choice: by F_ST=0.1 at the largest scale, the two weighted variants (free+two-step, fixed) reach substantially higher estimates than their unweighted counterparts, regardless of whether the intercept is free+two-step or fixed. Weighting and intercept-stabilization are two different knobs, not two readings of the same axis.

Consistent with the well-behaved scenarios: in the three scenarios without an f-mismatched reference panel, all six variants track each other far more closely (see the figures for 'No ref panel' and 'Ref panel, shared f') - this split is not a general property of these six estimators, it is specifically triggered by the f-mismatch failure mode.

Practical implication: if a reference panel's stratification structure is suspected to be mismatched with the target sample, checking whether a free-intercept, single-step LDSC fit collapses toward zero relative to the two-step or fixed-intercept fits is a cheap diagnostic - in a well-specified analysis, they should not disagree this dramatically. Note that a fixed intercept does not protect against inflation the way it protects against collapse - in this scenario it produced the single largest estimate of all six variants at every scale tested, from n=500 (0.96, vs. 0.28 for the next-highest variant) through the new n=10000 row (2.06, vs. 1.77) - though the gap to the weighted two-step variant narrows sharply with scale, from roughly 3.4× at n=500 to roughly 1.2× at n=10000, suggesting the two could plausibly cross over at larger scales still.